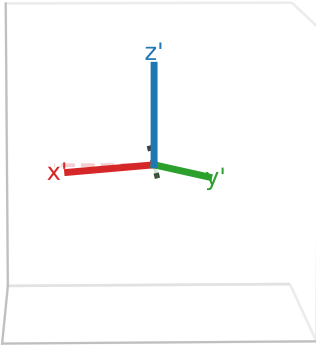
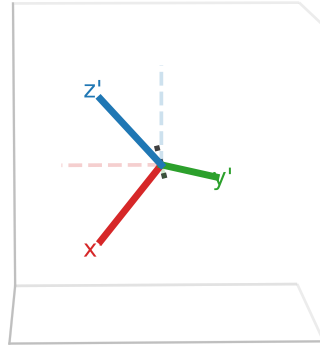


Gimbal lock occurs at roll = ± 90 deg in this convention

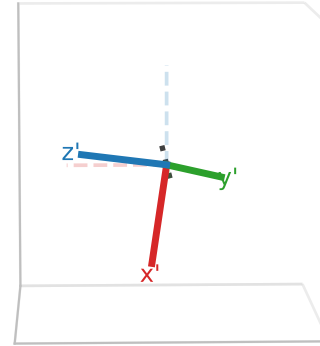
roll = 0 deg



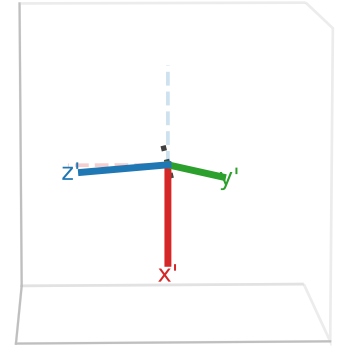
roll = 45 deg



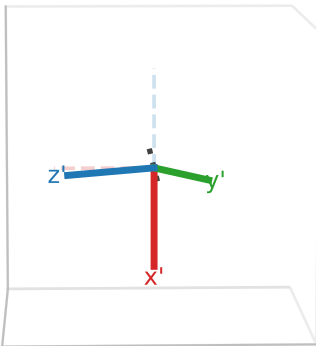
roll = 80 deg



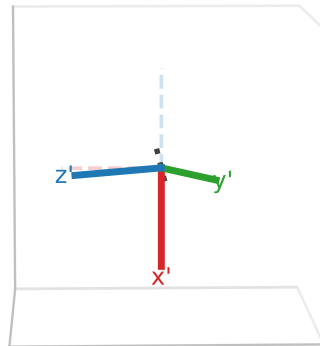
roll = 90 deg



roll = 90, yaw = 30, pitch = 0
(identical to its neighbour -- that is the lock)



roll = 90, yaw = 0, pitch = -30
(identical to its neighbour -- that is the lock)



Gimbal lock in the Chapter 2 convention

The singularity is at $\text{ROLL} = \pm 90$ deg (not pitch), because roll is the middle rotation about Y in $C = R_x(\theta) R_y(\phi) R_z(\psi)$.

```
(roll, pitch, yaw) = (90, 0, 30) -> recovered [ 90. -30.  0.]  
(roll, pitch, yaw) = (90, -30, 0) -> recovered [ 90. -30.  0.]
```

Both matrices agree to $3.1\text{e-}17$: the two inputs are the same attitude. Only pitch $\pm$ yaw survives, so the recovery pins $\text{yaw} = 0$ and folds the remainder into pitch.